

INCIDENT REPORT — INC-019

Equipment: Pump P-104 | Date: 08 October 2024 | Severity: HIGH

Reported by: Shift Supervisor — Unit 3 | Investigated by: Ravi Kumar, Senior Technician

1. Incident Description

At 03:45 hrs on 08 October 2024, continuous vibration monitoring alarmed for Pump P-104 at 7.8 mm/s. Operations attempted to reduce load but vibration climbed to 8.2 mm/s within 20 minutes. Pump was tripped at 04:05 hrs. Bearing temperature at trip was 89°C. No personnel injury. Production loss: 4 hours.

2. Root Cause Analysis

2.1 Immediate Cause: Complete bearing failure — both radial bearings seized. Grease completely carbonised. Rolling element damage visible on inspection.

2.2 Contributing Factor: Lubrication interval exceeded by approximately 100 hours. Last lubrication (WO-2422) was on 12 August 2024. Estimated operating hours at failure: approximately 600 hours since lubrication. OEM limit is 500 hours.

2.3 Root Cause: Lubrication schedule not triggered in the CMMS after WO-2422. The 500-hour lubrication reminder was not set, and no manual follow-up was performed. This is the same root cause as the 2022 and 2023 bearing failures on P-104 — a systemic CMMS scheduling failure, not an isolated incident.

3. Failure Pattern — Historical

This is the THIRD bearing failure on P-104 caused by lubrication interval exceedance. 2022 failure: interval exceeded by 80 hours. 2023 failure: interval exceeded by 65 hours. 2024 failure: interval exceeded by 100 hours. In all three cases, vibration exceeded 7.1 mm/s approximately 2–3 weeks before catastrophic bearing failure. The pattern is consistent and predictable.

4. Corrective Actions

4.1 Immediate: Replace both bearings (SKF 6205-2RS1/C3). Re-lubricate. Verify alignment. Return to service after vibration confirmed < 2.3 mm/s.

4.2 Short-term: Set CMMS automated alert at 450-hour lubrication interval for P-104. Do not rely on manual scheduling.

4.3 Long-term: Evaluate automatic lubrication system for P-104 given recurring failures. Estimated cost: ₹1,80,000. Estimated failure prevention value: ₹4,50,000/year.

5. Compliance Reference

This incident and corrective actions are documented per OISD-113 requirements for maintenance of rotating equipment. Factory Act Section 7A notification submitted to factory inspector on 09 October 2024.